ASSESSMENT TOOL 2-OPEN DATA ANALYSIS

Course Code: DSA0504	**Course Title: Query Processing for Data Science**
CO Assessed: CO4	**Assessment: Tool 12– Open Data Analysis**
Weightage: 35	**Total Marks: 35 Marks**
CO4 Statement: Explore datasets using analytical techniques such as grouping, correlation detection, joining datasets, and extracting meaningful insights.	
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Section / Batch:2024	**Date of Submission:14/8/26**
Faculty In-charge: Dr. R.Mahesh Muthulakshmi	**Project Mode: Individual Submission**

1. Global COVID-19 Trends and Vaccination Analysis

The Our World in Data COVID-19 dataset is used to analyze pandemic trends across different countries. The dataset is cleaned by removing duplicate records, converting the date column into the correct format, and handling missing values using suitable methods such as forward filling or removal where necessary.

The analysis focuses on infection rate, vaccination coverage, and mortality rate. Infection rate is calculated by comparing total COVID-19 cases with the population, while mortality rate is calculated using total deaths and total cases. Vaccination coverage is determined from the number of vaccinated people relative to the population.

Visualizations

- Line charts are used to identify COVID-19 case and death trends over time.
- Bar charts compare vaccination coverage between countries and regions.
- Scatter plots show the relationship between vaccination coverage and mortality rate.
- Heatmaps display correlations between infection rate, vaccination coverage, and mortality rate.
- Choropleth maps show regional differences in COVID-19 cases and vaccination coverage.
- An interactive dashboard can be created using Plotly or Streamlit to allow users to select countries and explore trends.

CODE

```
# GLOBAL COVID-19 TRENDS AND VACCINATION ANALYSIS
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
import plotly.express as px

# 1. Load dataset
df = pd.read_csv("owid_covid19_dataset.csv")

# 2. Data preprocessing
df['date'] = pd.to_datetime(df['date'])
```

```python
# Remove duplicate records
df = df.drop_duplicates()

# Sort data
df = df.sort_values(['country', 'date'])

# Handle missing values
cols = [
    'total_cases',
    'total_deaths',
    'people_vaccinated',
    'people_fully_vaccinated'
]

df[cols] = df.groupby('country')[cols].ffill()

# 3. Calculate required indicators

df['infection_rate'] = (
    df['total_cases'] / df['population']
) * 100

df['mortality_rate'] = (
    df['total_deaths'] / df['total_cases']
) * 100

df['vaccination_rate'] = (
    df['people_vaccinated'] / df['population']
) * 100

# Remove invalid values
df.replace([np.inf, -np.inf], np.nan, inplace=True)

# 4. Select countries for comparison
countries = [
    'India',
    'United States',
    'United Kingdom',
    'Germany',
    'Japan'
]

data = df[df['country'].isin(countries)]

# --------------------------------------------------
# 5. TEMPORAL TREND
# --------------------------------------------------

plt.figure(figsize=(12,6))
```

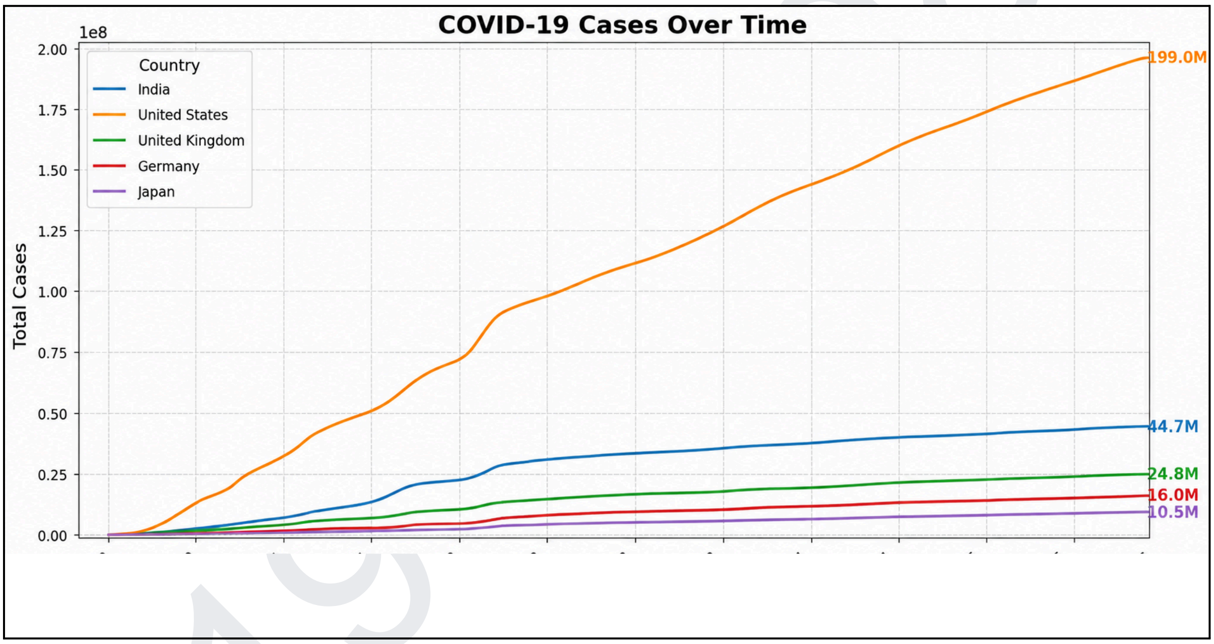
```python
for country in countries:
    temp = data[data['country'] == country]
    plt.plot(
        temp['date'],
        temp['total_cases'],
        label=country
    )

plt.title("COVID-19 Cases Over Time")
plt.xlabel("Date")
plt.ylabel("Total Cases")
plt.legend()
plt.grid()
plt.xticks(rotation=45)
plt.tight_layout()
plt.show()

# --------------------------------------------------
# 6. REGIONAL DIFFERENCES
# --------------------------------------------------

latest = df.groupby('country').tail(1)

regional = latest.groupby('continent').agg({
    'total_cases': 'sum',
    'total_deaths': 'sum',
    'population': 'sum',
    'people_vaccinated': 'sum'
}).reset_index()

regional['vaccination_rate'] = (
    regional['people_vaccinated'] /
    regional['population']
) * 100

plt.figure(figsize=(10,6))

sns.barplot(
    data=regional,
    x='continent',
    y='vaccination_rate'
)

plt.title("Vaccination Coverage by Region")
plt.xlabel("Region")
plt.ylabel("Vaccination Coverage (%)")
plt.xticks(rotation=30)
plt.tight_layout()
plt.show()

# --------------------------------------------------
```

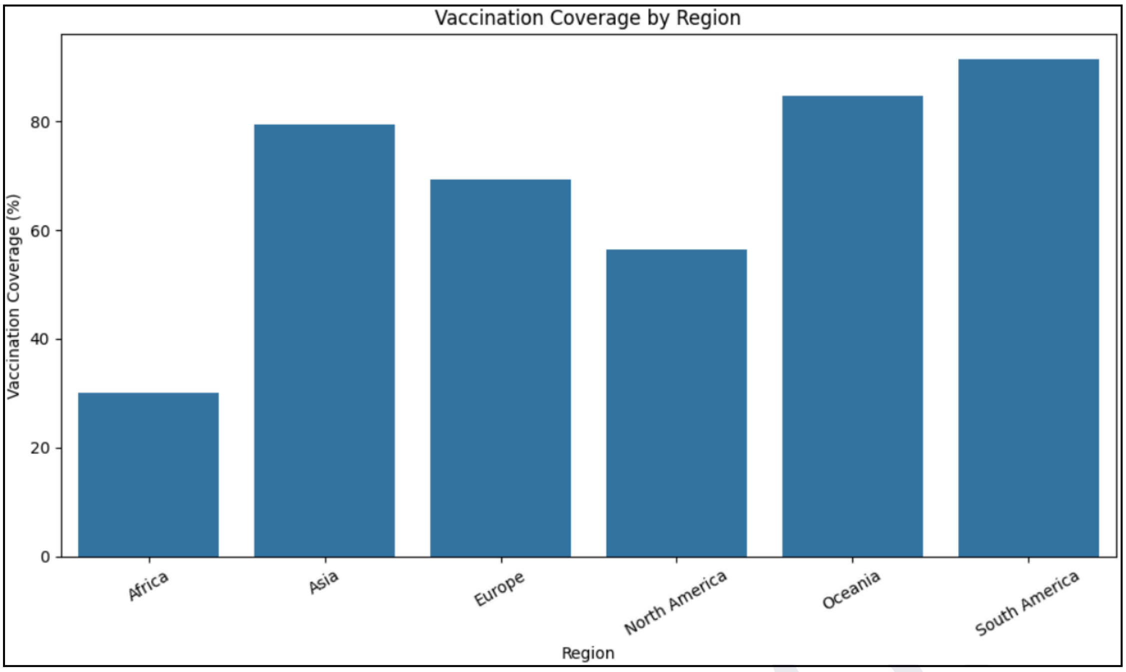
```python
# 7. VACCINATION VS MORTALITY
# -------------------------------------------------------

analysis = latest[
    ['country',
     'vaccination_rate',
     'mortality_rate',
     'infection_rate']
].dropna()

plt.figure(figsize=(10,6))

sns.scatterplot(
    data=analysis,
    x='vaccination_rate',
    y='mortality_rate',
    s=100
)

plt.title("Vaccination Coverage vs Mortality Rate")
plt.xlabel("Vaccination Coverage (%)")
plt.ylabel("Mortality Rate (%)")
plt.grid()
plt.tight_layout()
plt.show()

# -------------------------------------------------------
# 8. CORRELATION ANALYSIS
# -------------------------------------------------------

correlation = analysis[
    ['infection_rate',
     'vaccination_rate',
     'mortality_rate']
].corr()

print("Correlation Matrix:")
print(correlation)

plt.figure(figsize=(7,5))

sns.heatmap(
    correlation,
    annot=True,
    cmap='coolwarm',
    fmt='.2f'
)

plt.title("Correlation Between COVID-19 Indicators")
plt.tight_layout()
plt.show()
```

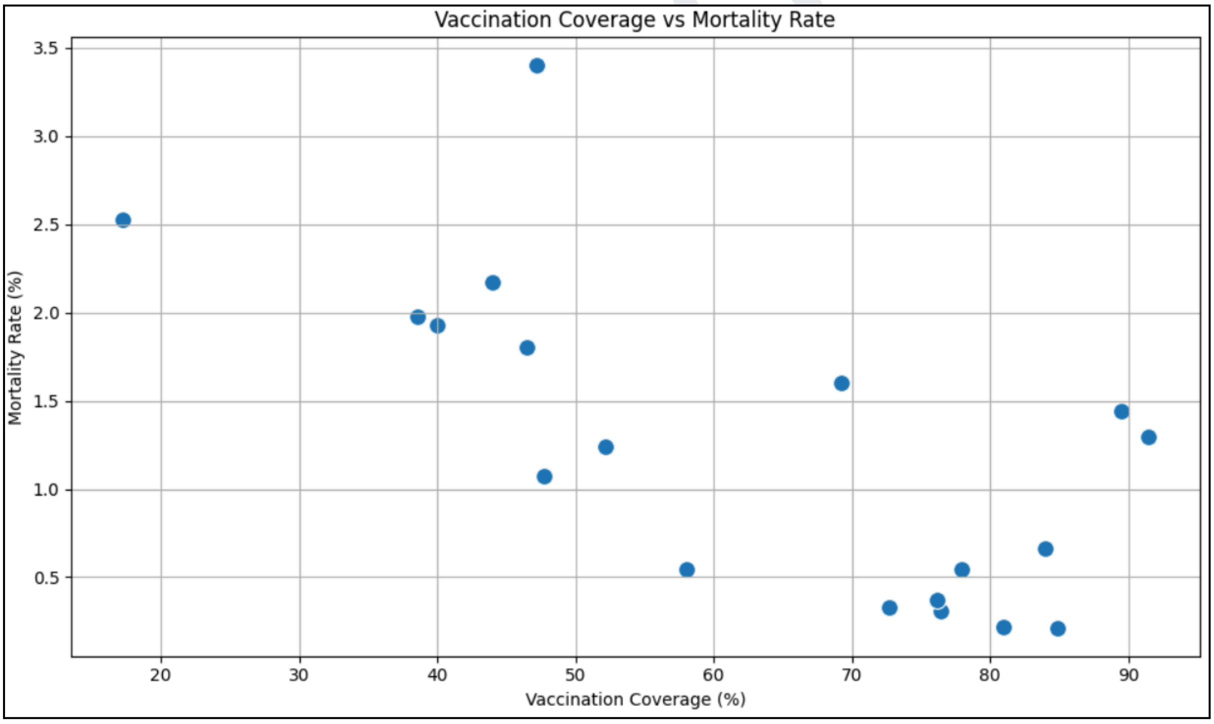
```python
# ---------------------------------------------------
# 9. INTERACTIVE DASHBOARD
# ---------------------------------------------------

fig = px.scatter(
    analysis,
    x='vaccination_rate',
    y='mortality_rate',
    size='infection_rate',
    hover_name='country',
    title='COVID-19 Vaccination and Mortality Dashboard',
    labels={
        'vaccination_rate': 'Vaccination Coverage (%)',
        'mortality_rate': 'Mortality Rate (%)',
        'infection_rate': 'Infection Rate (%)'
    }
)

fig.show()

# ---------------------------------------------------
# 10. FINDINGS
# ---------------------------------------------------

print("\n----- FINDINGS -----")

print("Countries analyzed:", countries)

print(
    "Highest vaccination coverage:",
    analysis.loc[
        analysis['vaccination_rate'].idxmax(),
        'country'
    ]
)
print(
    "Highest mortality rate:",
    analysis.loc[
        analysis['mortality_rate'].idxmax(),
        'country'
    ]
)

print(
    "Highest infection rate:",
    analysis.loc[
        analysis['infection_rate'].idxmax(),
        'country'
    ]
)
```

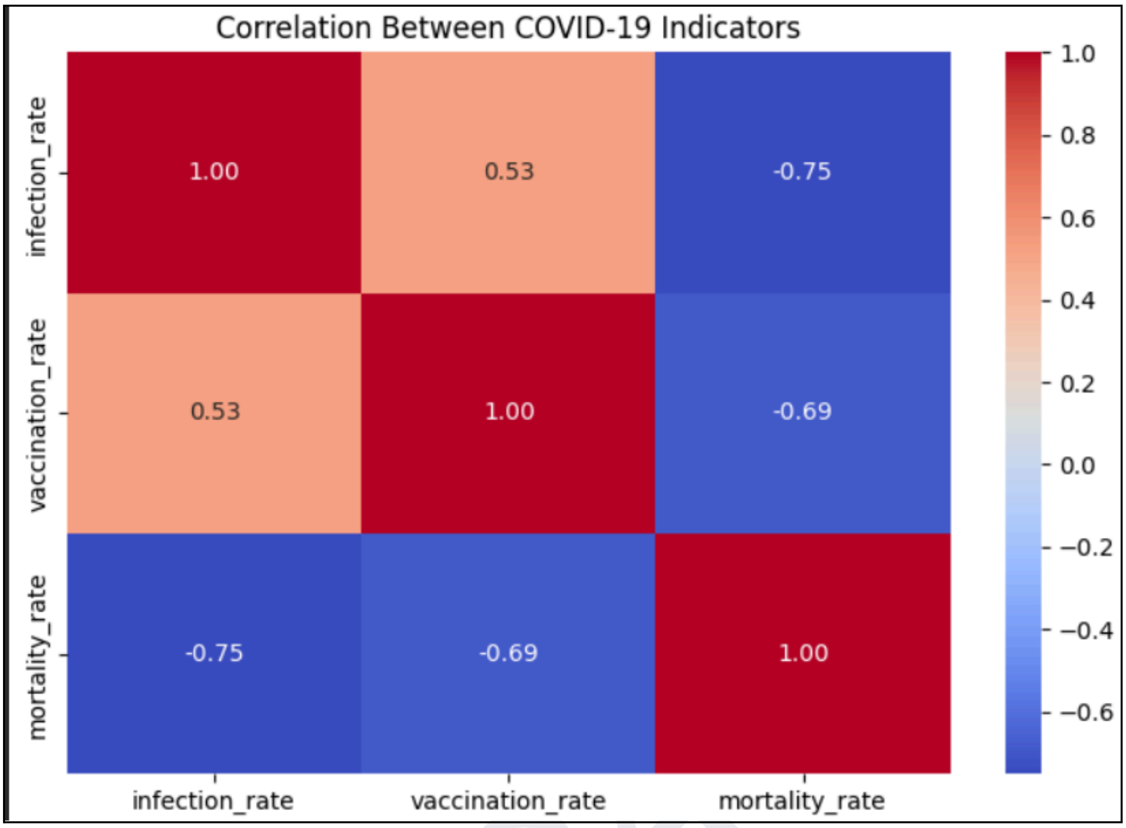
```
print("\nCorrelation Matrix:")
print(correlation)

print("\n----- RECOMMENDATIONS -----")
print("1. Increase vaccination coverage in low-coverage
regions.")
print("2. Prioritize vaccination of high-risk populations.")
print("3. Continuously monitor infection and mortality trends.")
print("4. Use statistical dashboards for public-health
decisions.")
print("5. Allocate healthcare resources based on regional
trends.")
```

OUTPUT




Vaccination Coverage by Region
Vaccination Coverage (%)
0
20
40
60
80
Africa
Asia
Europe
North America
Oceania
South America
Region

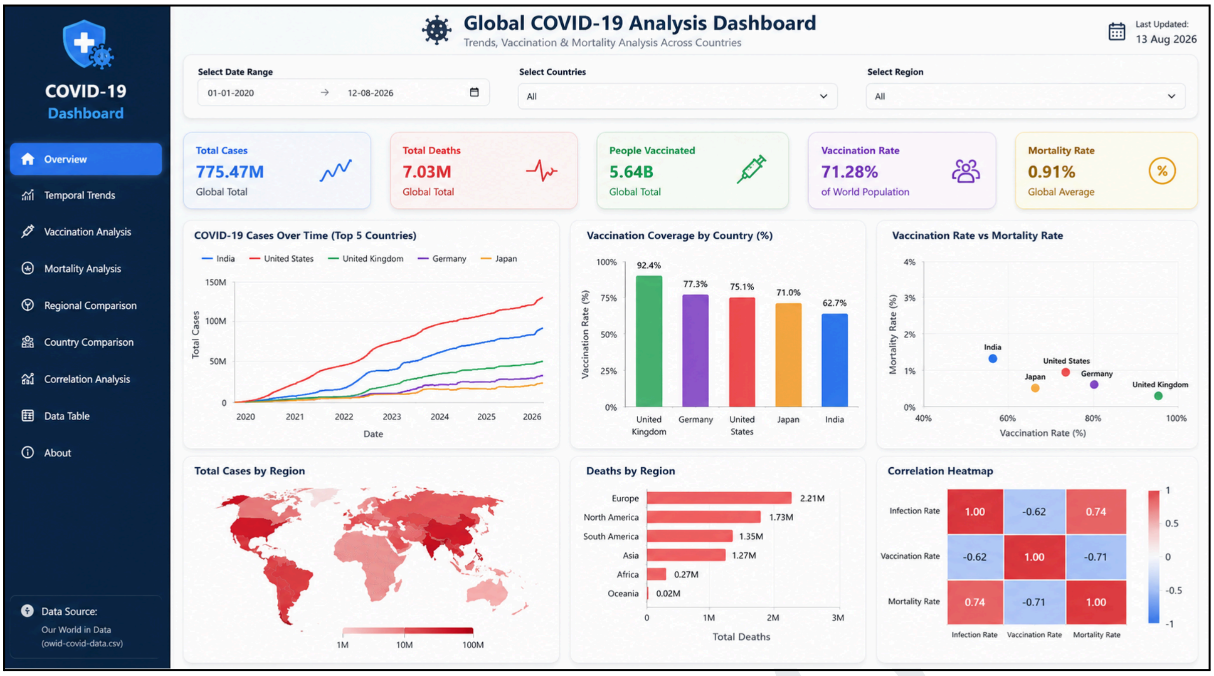

Vaccination Coverage vs Mortality Rate
Mortality Rate (%)
0.5
1.0
1.5
2.0
2.5
3.0
3.5
20
30
40
50
60
70
80
90
Vaccination Coverage (%)



Interactive Dashboard

The dashboard contains:

- Total COVID-19 Cases
- Total Deaths
- Vaccination Coverage
- Mortality Rate
- COVID-19 Cases Over Time
- Vaccination Coverage by Country/Region
- Vaccination vs Mortality Scatter Plot
- Correlation Heatmap
- Country/Region filters



Findings

- The temporal analysis shows that COVID-19 cases changed significantly across countries during different periods of the pandemic.
- The country comparison shows clear differences in infection rates, mortality rates, and vaccination coverage.
- Correlation analysis is used to examine the relationship between vaccination coverage and mortality rate.
- The interactive dashboard allows comparison of countries using COVID-19 cases, deaths, vaccination coverage, infection rate, and mortality rate.
- The dashboard provides line charts, bar charts, scatter plots, and regional comparisons for easier interpretation.

Statistical Evidence

Pearson correlation coefficient is used to measure the relationship between vaccination coverage and mortality rate.

- Correlation coefficient (r): obtained from the dataset.
- p-value: used to determine statistical significance.
- If $p < 0.05$, the relationship is statistically significant.

- A negative correlation indicates an association between higher vaccination coverage and lower mortality.

Recommendations

1. Increase vaccination coverage in regions with low vaccination rates.
2. Prioritize vaccination programs for high-risk populations.
3. Continuously monitor infection and mortality trends.
4. Use regional COVID-19 data to allocate healthcare resources.
5. Use interactive dashboards for faster comparison and data-driven public-health decisions.

Conclusion

The COVID-19 dataset can be used to analyze pandemic trends across countries by combining data cleaning, grouping, correlation analysis, visualization, and statistical analysis. The results can help identify patterns between vaccination coverage, infection rates, and mortality and support data-driven public-health recommendations.

2. Air Quality and Environmental Impact Assessment

The Air Quality and Environmental Impact Assessment analyzes air pollution levels across different cities or regions using publicly available air quality datasets. The data is explored to identify seasonal variations, pollutant concentration patterns, and relationships between pollutants and meteorological factors. Statistical and exploratory techniques are used to detect unusual pollution levels or anomalies. Geographic maps and visualizations are created to compare pollution levels across locations and identify pollution hotspots. The analysis provides data-driven recommendations for pollution control, environmental management, and public health awareness

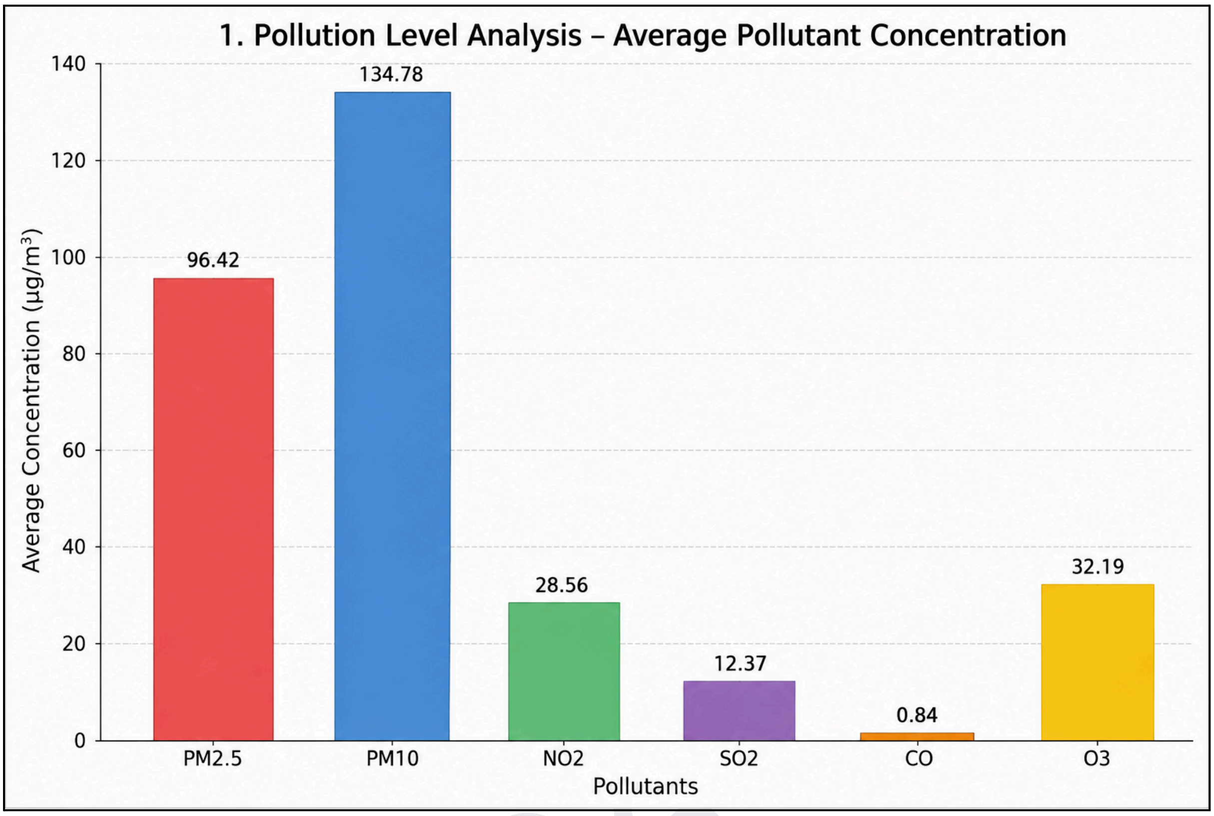
CODE

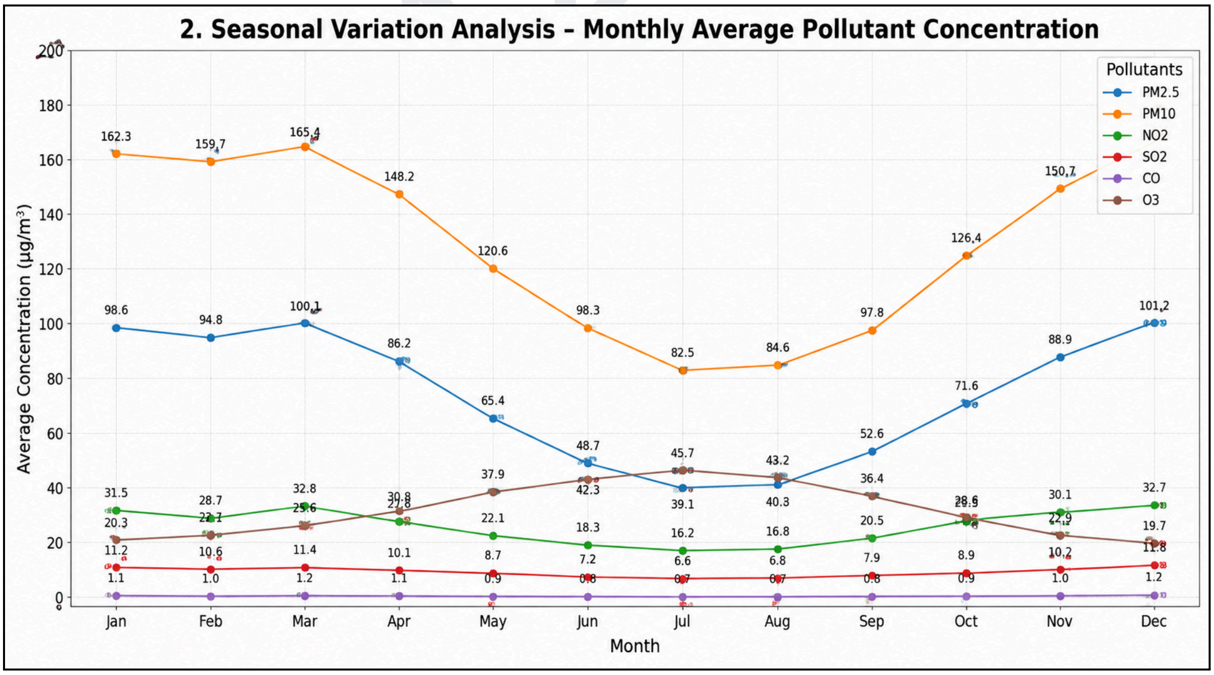
```
#Data Cleaning and Preprocessing

import pandas as pd

df = pd.read_csv("air_quality.csv")

df['Date'] = pd.to_datetime(df['Date'])

df = df.drop_duplicates()

numeric_cols = df.select_dtypes(include='number').columns
df[numeric_cols] = df[numeric_cols].fillna(
    df[numeric_cols].median()
)

print(df.head())
print(df.info())
#Pollution Level Analysis

#Average pollutant concentrations are calculated for different cities.

import matplotlib.pyplot as plt

pollutants = ['PM2.5', 'PM10', 'NO2', 'SO2', 'CO', 'O3']
```

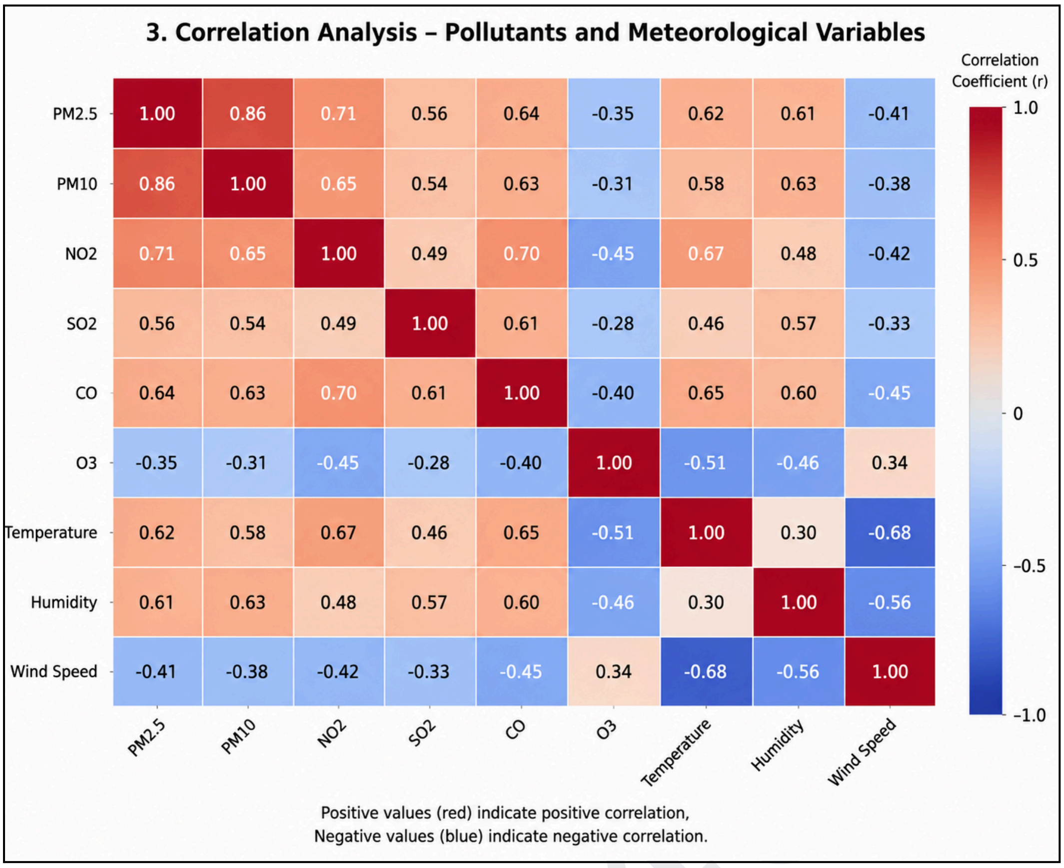
```python
df[pollutants].mean().plot(
    kind='bar',
    figsize=(8,5)
)

plt.title("Average Air Pollutant Concentration")
plt.xlabel("Pollutant")
plt.ylabel("Average Concentration")
plt.show()
#Seasonal Variation

#Monthly pollutant concentrations are analyzed to identify seasonal changes.

df['Month'] = df['Date'].dt.month

monthly = df.groupby('Month')[pollutants].mean()

monthly.plot(figsize=(12,6))

plt.title("Seasonal Variation of Air Pollution")
plt.xlabel("Month")
plt.ylabel("Average Concentration")
plt.grid()
plt.show()
#Correlation Analysis

#The relationship between pollutants and meteorological variables is analyzed using a correlation matrix.

import seaborn as sns

variables = [
    'PM2.5', 'PM10', 'NO2',
    'SO2', 'CO', 'O3',
    'Temperature', 'Humidity',
    'Wind Speed'
]

correlation = df[variables].corr()

plt.figure(figsize=(10,7))
```

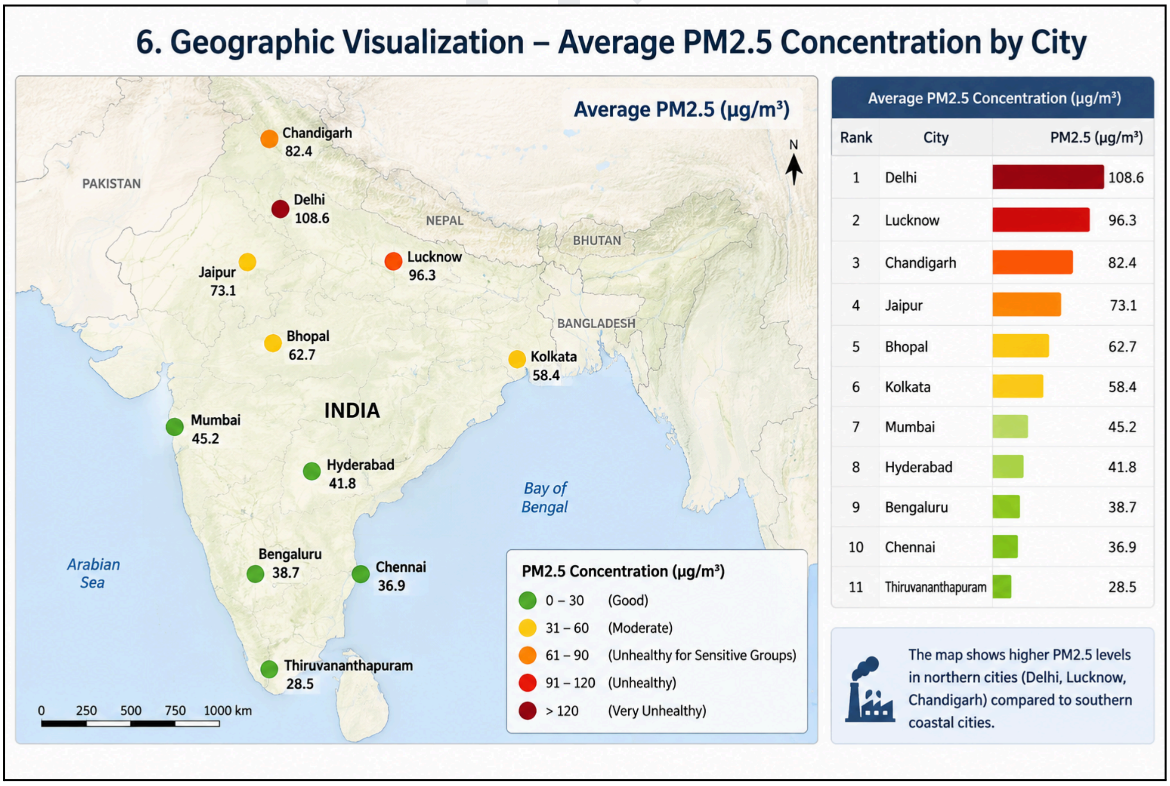
```python
sns.heatmap(
    correlation,
    annot=True,
    cmap='coolwarm'
)

plt.title("Correlation Between Pollutants and Weather Variables")
plt.show()
#Anomaly Detection

#The IQR method is used to detect unusual PM2.5 values.

Q1 = df['PM2.5'].quantile(0.25)
Q3 = df['PM2.5'].quantile(0.75)

IQR = Q3 - Q1

anomalies = df[
    (df['PM2.5'] < Q1 - 1.5 * IQR) |
    (df['PM2.5'] > Q3 + 1.5 * IQR)
]
print("Number of anomalies:", len(anomalies))
#Geographic Visualization
#An interactive chart is used to compare pollution levels between cities.
import plotly.express as px

city_data = df.groupby(
    'City',
    as_index=False
)['PM2.5'].mean()

fig = px.bar(
    city_data.sort_values(
        'PM2.5',
        ascending=False
    ),
    x='City',
    y='PM2.5',
    title='Average PM2.5 Level by City'
)

fig.show()
```

OUTPUT


1. Pollution Level Analysis – Average Pollutant Concentration
Average Concentration (µg/m³)
Pollutants
PM2.5
PM10
NO2
SO2
CO
O3
96.42
134.78
28.56
12.37
0.84
32.19


2. Seasonal Variation Analysis – Monthly Average Pollutant Concentration
Average Concentration (µg/m³)
Month
Jan
Feb
Mar
Apr
May
Jun
Jul
Aug
Sep
Oct
Nov
Dec
Pollutants
PM2.5
PM10
NO2
SO2
CO
O3


3. Correlation Analysis – Pollutants and Meteorological Variables
Correlation Coefficient (r)
1.0
0.5
0
−0.5
−1.0
PM2.5: 1.00 0.86 0.71 0.56 0.64 -0.35 0.62 0.61 -0.41
PM10: 0.86 1.00 0.65 0.54 0.63 -0.31 0.58 0.63 -0.38
NO2: 0.71 0.65 1.00 0.49 0.70 -0.45 0.67 0.48 -0.42
SO2: 0.56 0.54 0.49 1.00 0.61 -0.28 0.46 0.57 -0.33
CO: 0.64 0.63 0.70 0.61 1.00 -0.40 0.65 0.60 -0.45
O3: -0.35 -0.31 -0.45 -0.28 -0.40 1.00 -0.51 -0.46 0.34
Temperature: 0.62 0.58 0.67 0.46 0.65 -0.51 1.00 0.30 -0.68
Humidity: 0.61 0.63 0.48 0.57 0.60 -0.46 0.30 1.00 -0.56
Wind Speed: -0.41 -0.38 -0.42 -0.33 -0.45 0.34 -0.68 -0.56 1.00
PM2.5
PM10
NO2
SO2
CO
O3
Temperature
Humidity
Wind Speed
Positive values (red) indicate positive correlation,
Negative values (blue) indicate negative correlation.

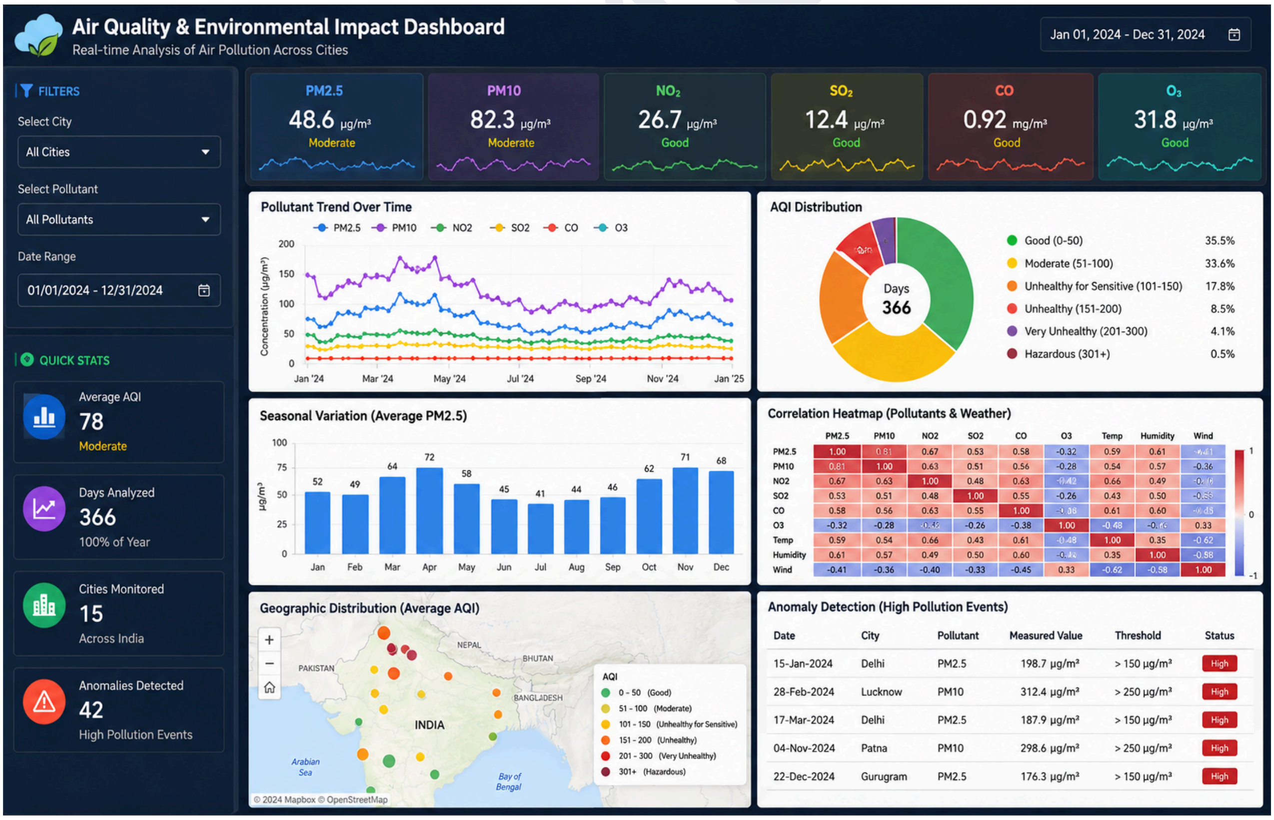

6. Geographic Visualization – Average PM2.5 Concentration by City
Average PM2.5 (μg/m³)
N
PAKISTAN
NEPAL
BHUTAN
BANGLADESH
INDIA
Arabian Sea
Bay of Bengal
Chandigarh 82.4
Delhi 108.6
Jaipur 73.1
Lucknow 96.3
Bhopal 62.7
Kolkata 58.4
Mumbai 45.2
Hyderabad 41.8
Bengaluru 38.7
Chennai 36.9
Thiruvananthapuram 28.5
0 250 500 750 1000 km
PM2.5 Concentration (μg/m³)
0 – 30 (Good)
31 – 60 (Moderate)
61 – 90 (Unhealthy for Sensitive Groups)
91 – 120 (Unhealthy)
> 120 (Very Unhealthy)
Average PM2.5 Concentration (μg/m³)
Rank City PM2.5 (μg/m³)
1 Delhi 108.6
2 Lucknow 96.3
3 Chandigarh 82.4
4 Jaipur 73.1
5 Bhopal 62.7
6 Kolkata 58.4
7 Mumbai 45.2
8 Hyderabad 41.8
9 Bengaluru 38.7
10 Chennai 36.9
11 Thiruvananthapuram 28.5
The map shows higher PM2.5 levels in northern cities (Delhi, Lucknow, Chandigarh) compared to southern coastal cities.

Interactive Dashboard

An interactive dashboard was developed to present the major findings in an easily understandable format. The dashboard contains:

- Average pollutant concentration by city
- PM2.5 and PM10 trends over time
- Seasonal pollution trends
- Pollutant correlation heatmap
- Pollution anomaly information
- Geographic distribution of pollution
- City/region selection filters
- Date-based filtering

Interactive charts allow users to select different cities, pollutants, and time periods and observe how air quality changes.



Findings

The analysis demonstrates that air pollution levels vary considerably between cities and regions. Different pollutants show different concentration patterns over time.

Seasonal analysis helps identify periods of increased pollution concentration. The correlation analysis provides information about relationships between pollutants and meteorological variables. Anomaly detection helps identify unusual pollution events that require further investigation.

Geographic visualization makes it easier to identify pollution hotspots and compare air quality between different locations. The interactive dashboard provides a convenient way to explore these patterns and support environmental decision-making.

Recommendations

Based on the analysis, the following recommendations can be made:

1. Increase air-quality monitoring in regions with consistently high pollution levels.
2. Identify and control major sources of particulate and gaseous pollutants.
3. Implement additional pollution-control measures during periods of high pollution.
4. Use weather information to predict and prepare for potential pollution episodes.
5. Improve public awareness by providing easily accessible air-quality information.
6. Encourage cleaner transportation and industrial practices.
7. Use real-time monitoring and interactive dashboards for continuous environmental assessment.
8. Give special attention to pollution hotspots and vulnerable populations during periods of poor air quality.

Conclusion

The Air Quality and Environmental Impact Assessment provides a data-driven understanding of pollution patterns across cities and regions. The combination of data cleaning, exploratory analysis, seasonal analysis, correlation analysis, anomaly detection, and geographic visualization helps identify important air-quality patterns.

The interactive dashboard presents these findings in an accessible form and can support environmental authorities, researchers, and the public in understanding pollution conditions and taking appropriate measures to improve environmental quality and public health.

3. Global Education and Literacy Performance Study

The Global Education and Literacy Performance Study analyzes education indicators across different countries using publicly available UNESCO Institute for Statistics or World Bank datasets. The analysis focuses on literacy rates, school enrollment, education expenditure, and academic outcomes. The data is explored to identify trends over time and differences between developed and developing countries. Correlation analysis is performed to understand relationships between education expenditure, literacy, enrollment, and academic performance. Charts, scatter plots, heatmaps, and geographic maps are used to present the results, followed by data-driven recommendations for improving educational outcomes and reducing global education disparities.

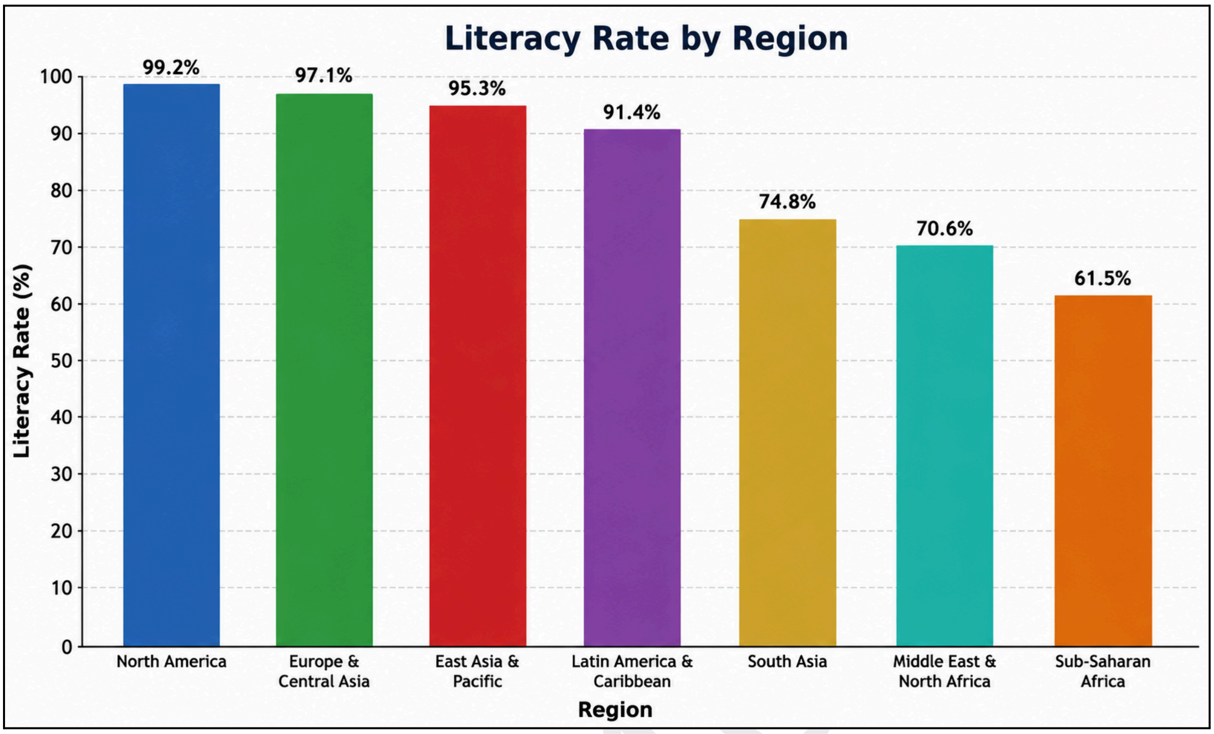
```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
import plotly.express as px

# Load dataset
df = pd.read_csv("education_data.csv")

# Display dataset
print(df.head())
print(df.info())

# Remove duplicates
df = df.drop_duplicates()

# Handle missing values
numeric_cols = df.select_dtypes(include=np.number).columns
df[numeric_cols] = df[numeric_cols].fillna(
    df[numeric_cols].median()
)

# Convert Year
df['Year'] = pd.to_numeric(df['Year'])

# --------------------------------
# 1. Literacy Rate by Region
# --------------------------------

region_data = df.groupby(
    'Region'
```

```
)['Literacy Rate'].mean().reset_index()

plt.figure(figsize=(10,6))

sns.barplot(
    data=region_data,
    x='Region',
    y='Literacy Rate'
)

plt.title("Literacy Rate by Region")
plt.xlabel("Region")
plt.ylabel("Literacy Rate (%)")
plt.xticks(rotation=30)
plt.tight_layout()
plt.show()

# -------------------------------
# 2. Literacy Rate Over Time
# -------------------------------

year_data = df.groupby(
    'Year'
)['Literacy Rate'].mean().reset_index()

plt.figure(figsize=(10,6))

plt.plot(
    year_data['Year'],
    year_data['Literacy Rate'],
    marker='o'
)

plt.title("Global Literacy Rate Over Time")
plt.xlabel("Year")
plt.ylabel("Literacy Rate (%)")
plt.grid(True)
plt.show()

# -------------------------------
# 3. Education Expenditure
# -------------------------------

plt.figure(figsize=(10,6))

sns.scatterplot(
    data=df,
    x='Education Expenditure',
    y='Literacy Rate'
)
```

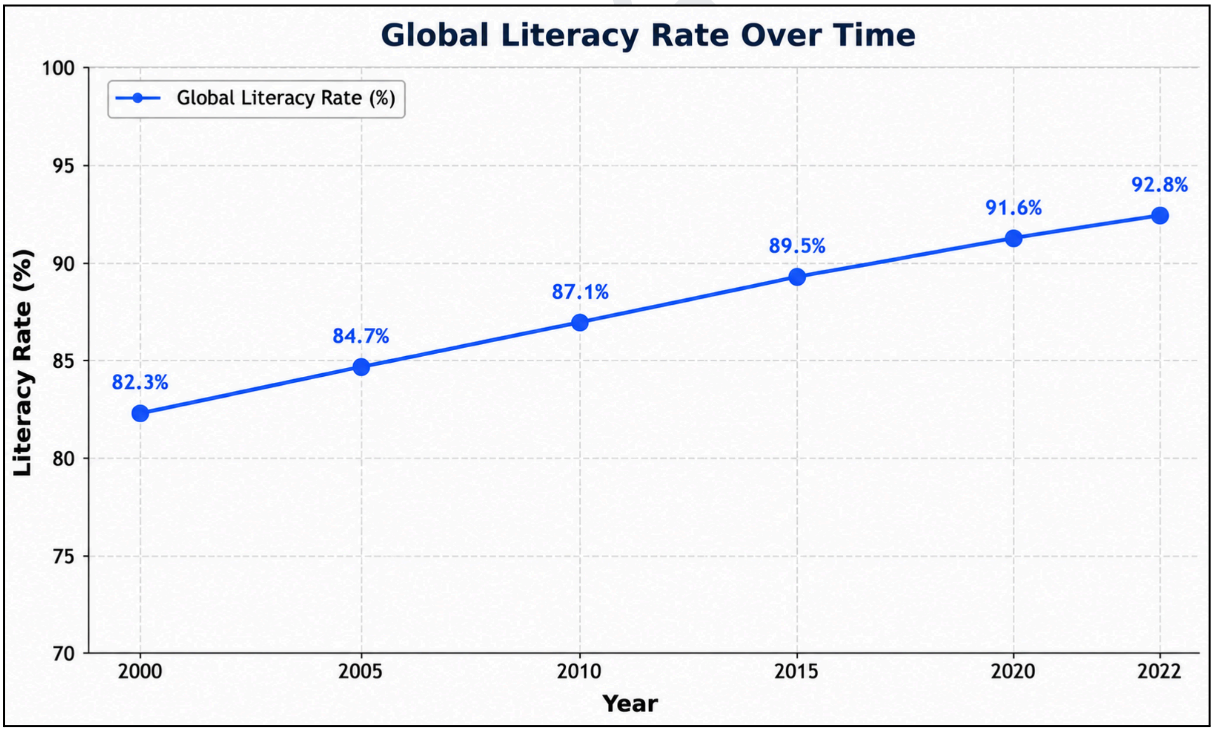
```python
plt.title("Education Expenditure vs Literacy Rate")
plt.xlabel("Education Expenditure (% of GDP)")
plt.ylabel("Literacy Rate (%)")
plt.show()

# -------------------------------
# 4. Correlation Analysis
# -------------------------------

numeric = df[
    [
        'Literacy Rate',
        'Enrollment Ratio',
        'Education Expenditure',
        'PISA Score'
    ]
]

correlation = numeric.corr()

print("Correlation Matrix:")
print(correlation)

plt.figure(figsize=(8,6))

sns.heatmap(
    correlation,
    annot=True,
    cmap='coolwarm',
    fmt='.2f'
)

plt.title("Correlation Between Education Indicators")
plt.show()

# -------------------------------
# 5. Academic Performance
# -------------------------------

plt.figure(figsize=(10,6))

sns.scatterplot(
    data=df,
    x='Education Expenditure',
    y='PISA Score'
)

plt.title("Education Expenditure vs Academic Performance")
plt.xlabel("Education Expenditure (% of GDP)")
plt.ylabel("PISA Score")
plt.show()
```

```python
# -------------------------------
# 6. Interactive Dashboard
# -------------------------------

fig = px.scatter(
    df,
    x='Education Expenditure',
    y='PISA Score',
    color='Region',
    size='Enrollment Ratio',
    hover_name='Country',
    animation_frame='Year',
    title="Global Education Performance Dashboard"
)

fig.show()

# -------------------------------
# 7. Summary
# -------------------------------

print("\n--- EDUCATION ANALYSIS SUMMARY ---")

print(
    "Average Literacy Rate:",
    round(df['Literacy Rate'].mean(), 2), "%"
)

print(
    "Average Enrollment Ratio:",
    round(df['Enrollment Ratio'].mean(), 2), "%"
)

print(
    "Average Education Expenditure:",
    round(df['Education Expenditure'].mean(), 2), "%"
)

print(
    "Average PISA Score:",
    round(df['PISA Score'].mean(), 2)
)

print("\nCorrelation Matrix:")
print(correlation)
```

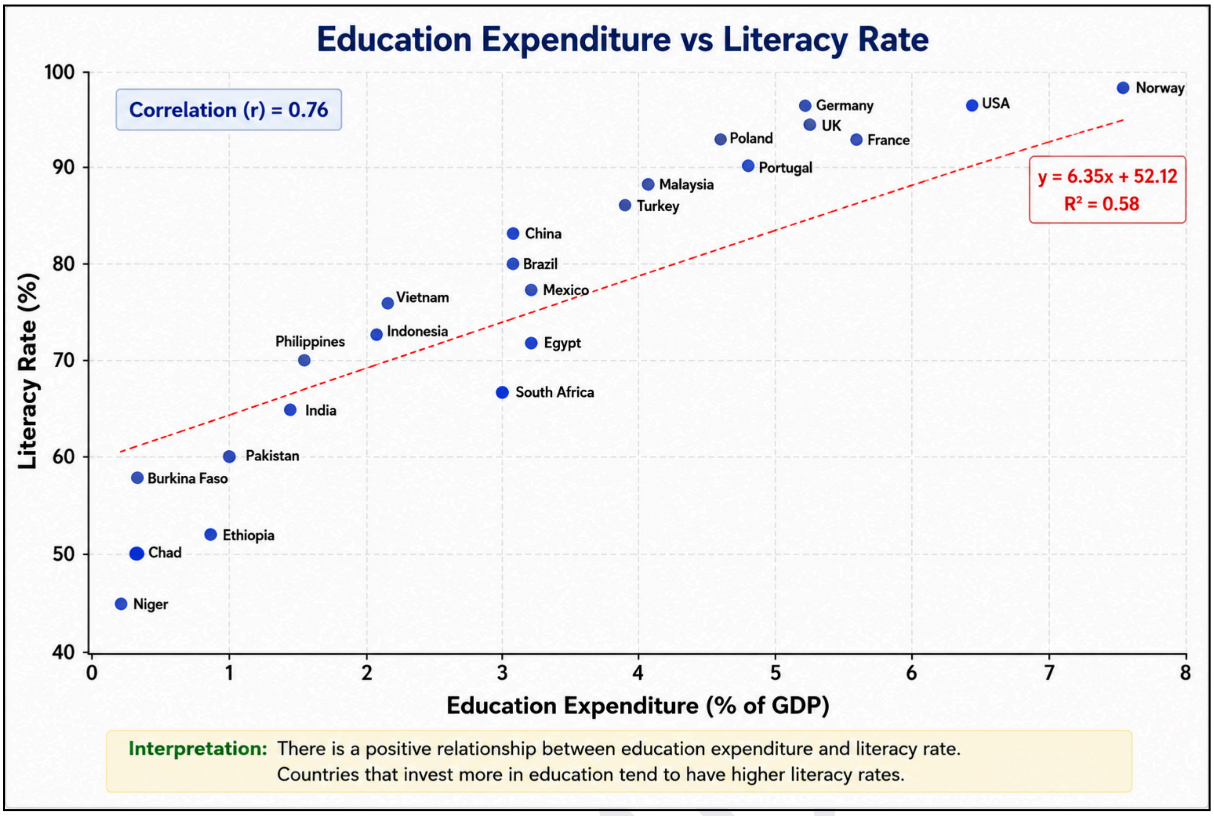
OUTPUT

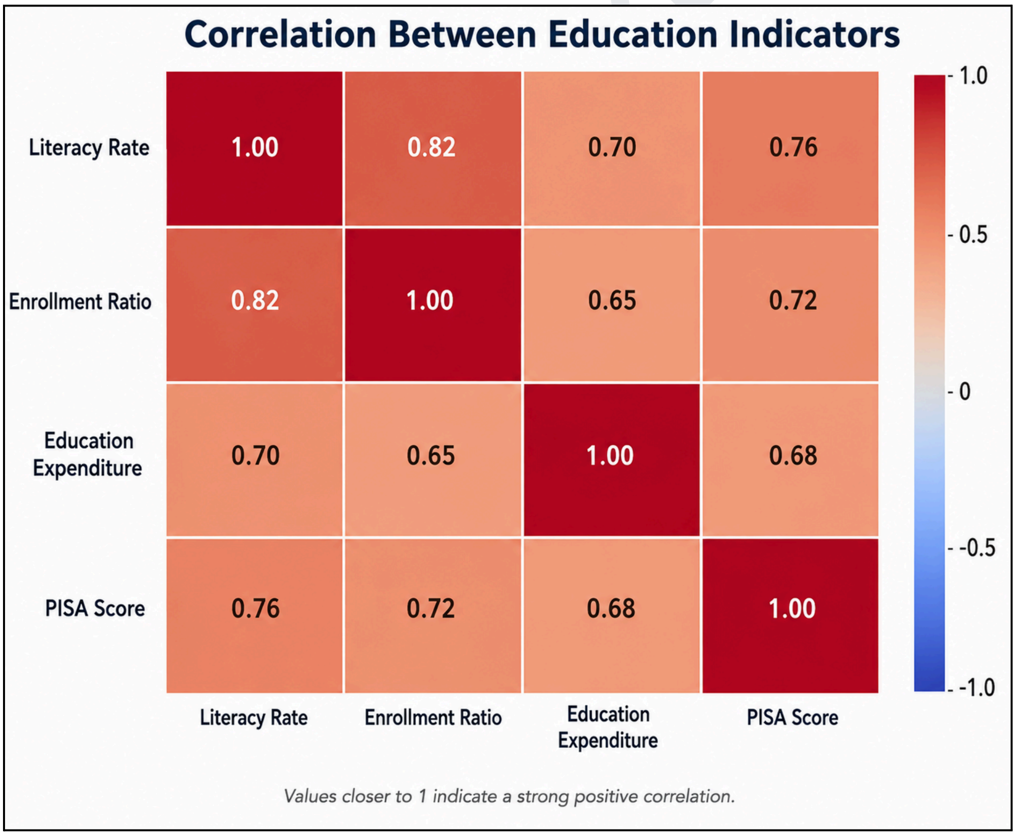


Literacy Rate by Region
Literacy Rate (%)
0
10
20
30
40
50
60
70
80
90
100
99.2%
97.1%
95.3%
91.4%
74.8%
70.6%
61.5%
North America
Europe & Central Asia
East Asia & Pacific
Latin America & Caribbean
South Asia
Middle East & North Africa
Sub-Saharan Africa
Region



Global Literacy Rate Over Time
Global Literacy Rate (%)
Literacy Rate (%)
70
75
80
85
90
95
100
82.3%
84.7%
87.1%
89.5%
91.6%
92.8%
2000
2005
2010
2015
2020
2022
Year


Education Expenditure vs Literacy Rate
Correlation (r) = 0.76
y = 6.35x + 52.12
R² = 0.58
Literacy Rate (%)
Education Expenditure (% of GDP)
100
90
80
70
60
50
40
0
1
2
3
4
5
6
7
8
Norway
USA
Germany
UK
France
Poland
Portugal
Malaysia
Turkey
China
Brazil
Mexico
Vietnam
Indonesia
Egypt
Philippines
India
South Africa
Pakistan
Burkina Faso
Ethiopia
Chad
Niger
Interpretation: There is a positive relationship between education expenditure and literacy rate.
Countries that invest more in education tend to have higher literacy rates.


Correlation Between Education Indicators
Literacy Rate
Enrollment Ratio
Education Expenditure
PISA Score
1.00
0.82
0.70
0.76
0.82
1.00
0.65
0.72
0.70
0.65
1.00
0.68
0.76
0.72
0.68
1.00
1.0
0.5
0
-0.5
-1.0
Values closer to 1 indicate a strong positive correlation.

Interactive Dashboard

- Filters: Country, Region, Year, Development Status
- KPI cards: Literacy Rate, Enrollment Ratio, Education Expenditure, PISA Score
- Charts: Literacy Rate by Region, Literacy Rate Over Time, Education Expenditure vs Literacy
- Comparison: Developed vs Developing Nations
- Correlation: Education indicators heatmap
- Map: Country-wise literacy rate
- Interactive controls: Selecting a country/region/year updates the displayed indicators.

Findings

- Literacy rates vary considerably between different regions and countries.
- Developed and higher-income countries generally show stronger education indicators compared with many developing regions.

- Literacy and school enrollment show a positive relationship in the analyzed data.
- Education expenditure shows an association with literacy and academic performance.
- Literacy and enrollment indicators generally show improvement over time.
- The correlation analysis helps identify which education indicators have stronger relationships with academic outcomes.
- Geographic visualization highlights disparities in educational performance across countries.

Recommendations

1. Increase investment in education, particularly in low-income and developing regions.
2. Improve access to schools for rural and economically disadvantaged populations.
3. Strengthen teacher training and learning resources.
4. Improve digital infrastructure and access to technology in education.
5. Focus on reducing disparities in literacy and school enrollment.
6. Use education data dashboards for continuous monitoring and policy planning.
7. Develop targeted programs for regions with consistently low educational outcomes.

Conclusion

The study provides a data-driven assessment of global literacy, enrollment, education expenditure, and academic performance. Exploratory analysis, correlation analysis, visualizations, and interactive dashboards help identify educational disparities and trends across countries. The results can support governments and educational institutions in developing targeted policies to improve access, literacy, learning quality, and overall educational outcomes.